MULTIPLAYER TURN-BASED GAME ENGINE USING JAVA

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PROBLEM:

LACK OF REUSABLE ARCHITECTURE IN TURN-BASED GAMES

TARGET USER:

STUDENTS AND BEGINNER DEVELOPERS

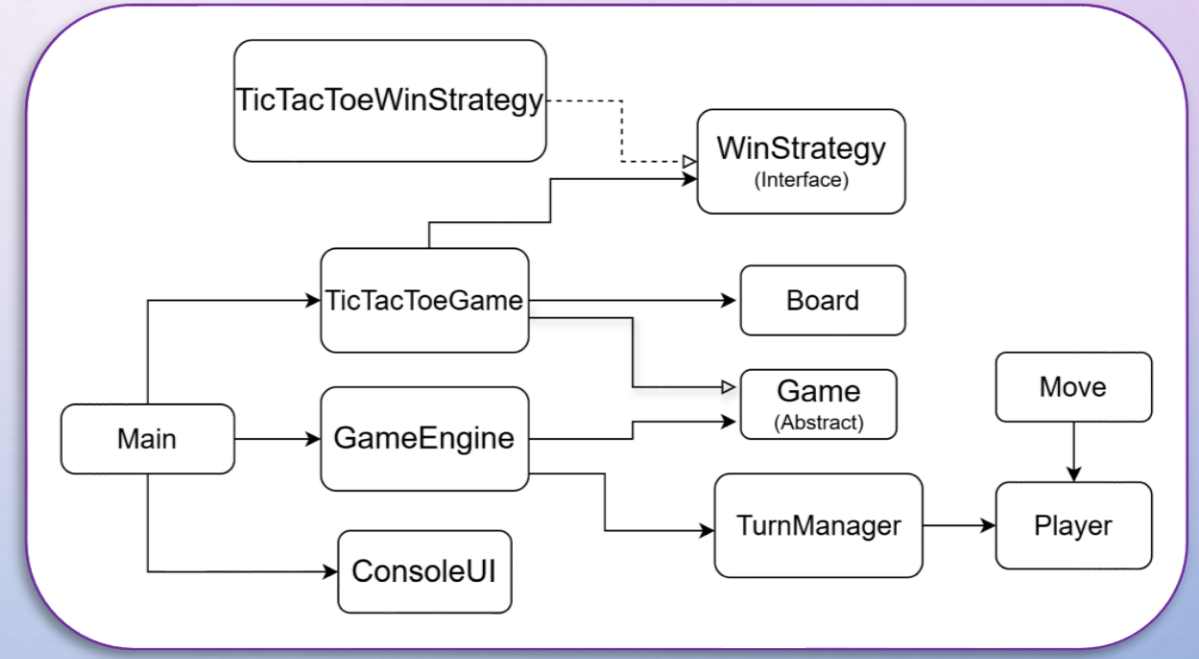
System Design

ARCHITECTURE:

UI → ENGINE → GAME → STRATEGY → MODEL

KEY CLASSES:

- **GAME**
- **TICTACTOEGAME**
- **GAMEENGINE**
- **TURNMANAGER**
- **PLAYER**



PACKAGES:

APP, ENGINE, MODEL, GAMES, STRATEGY, UI

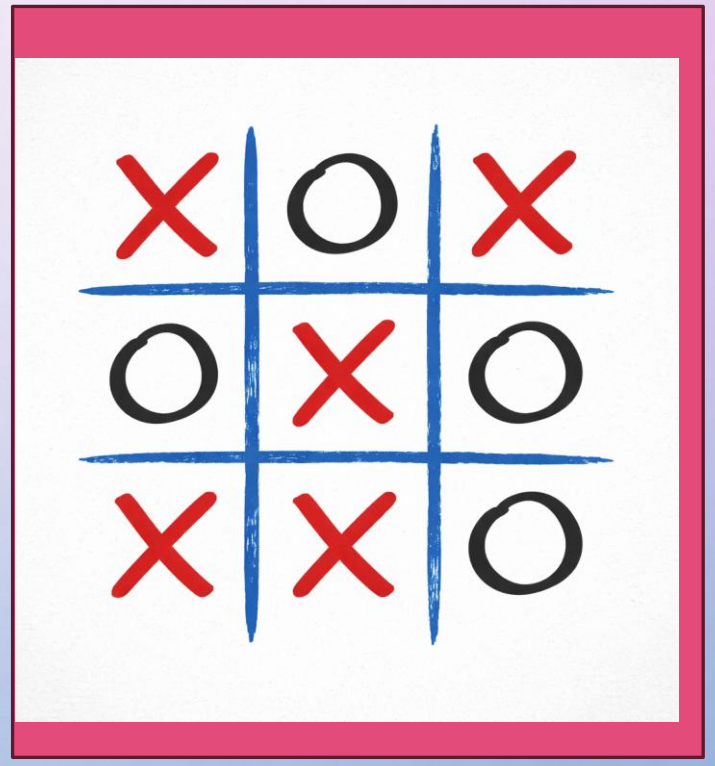
OOP Concepts Used

- ABSTRACTION → GAME ABSTRACT CLASS
- INHERITANCE → TICTACTOEGAME EXTENDS GAME
- POLYMORPHISM → OVERRIDDEN METHODS (CHECKWINNER)
- ENCAPSULATION → PLAYER CLASS (PRIVATE DATA)
- DESIGN PATTERN → STRATEGY PATTERN (WINNER LOGIC)

TECHNICAL HIGHLIGHT

STRATEGY PATTERN

- USED FOR WINNER DETECTION LOGIC
- SEPARATES GAME RULES FROM ENGINE
- ALLOWS EASY ADDITION OF NEW GAMES



Live Demo

- GAME EXECUTION
- VALID MOVE
- INVALID MOVE (ERROR HANDLING)
- WINNER DETECTION
- DRAW CONDITION